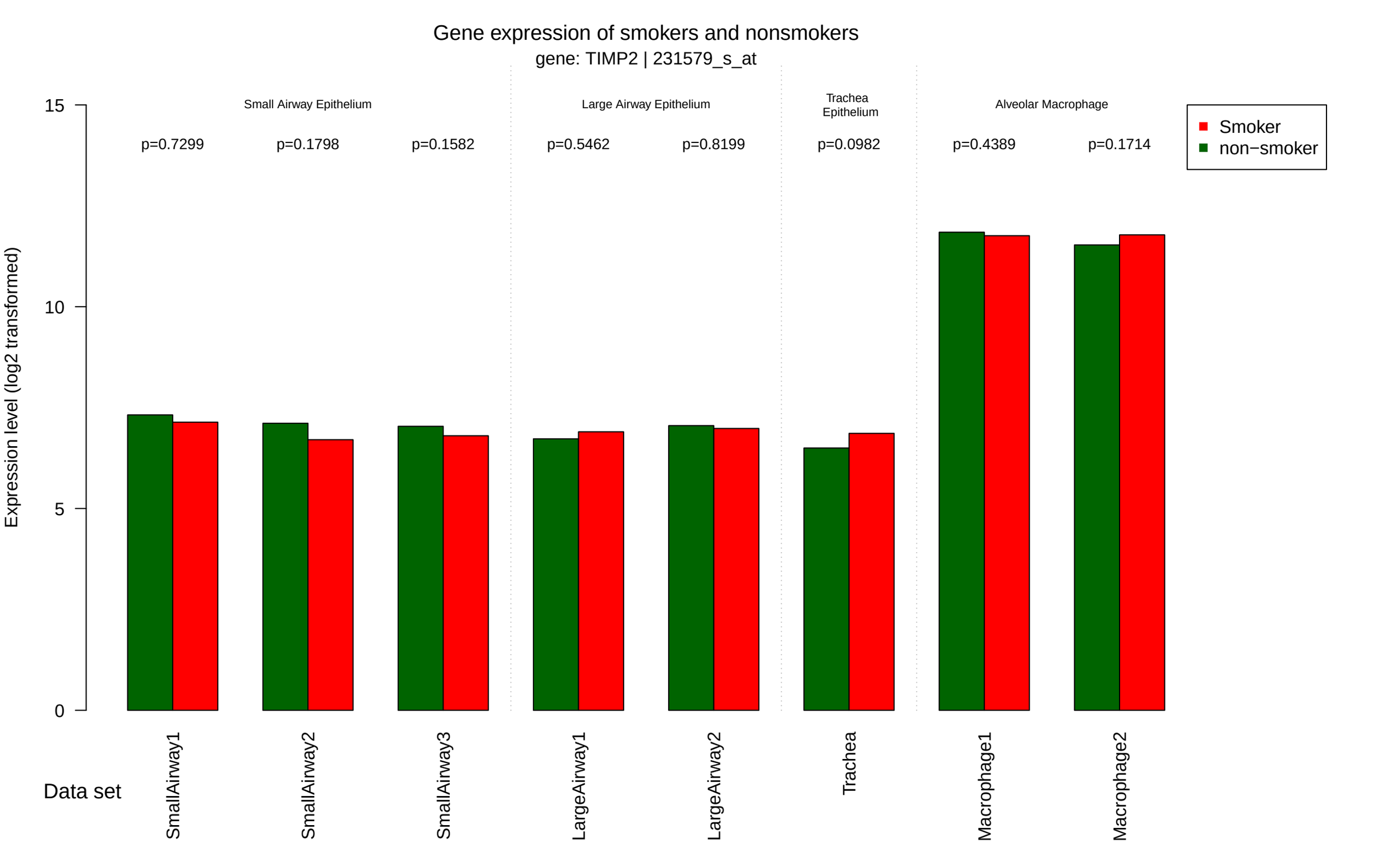
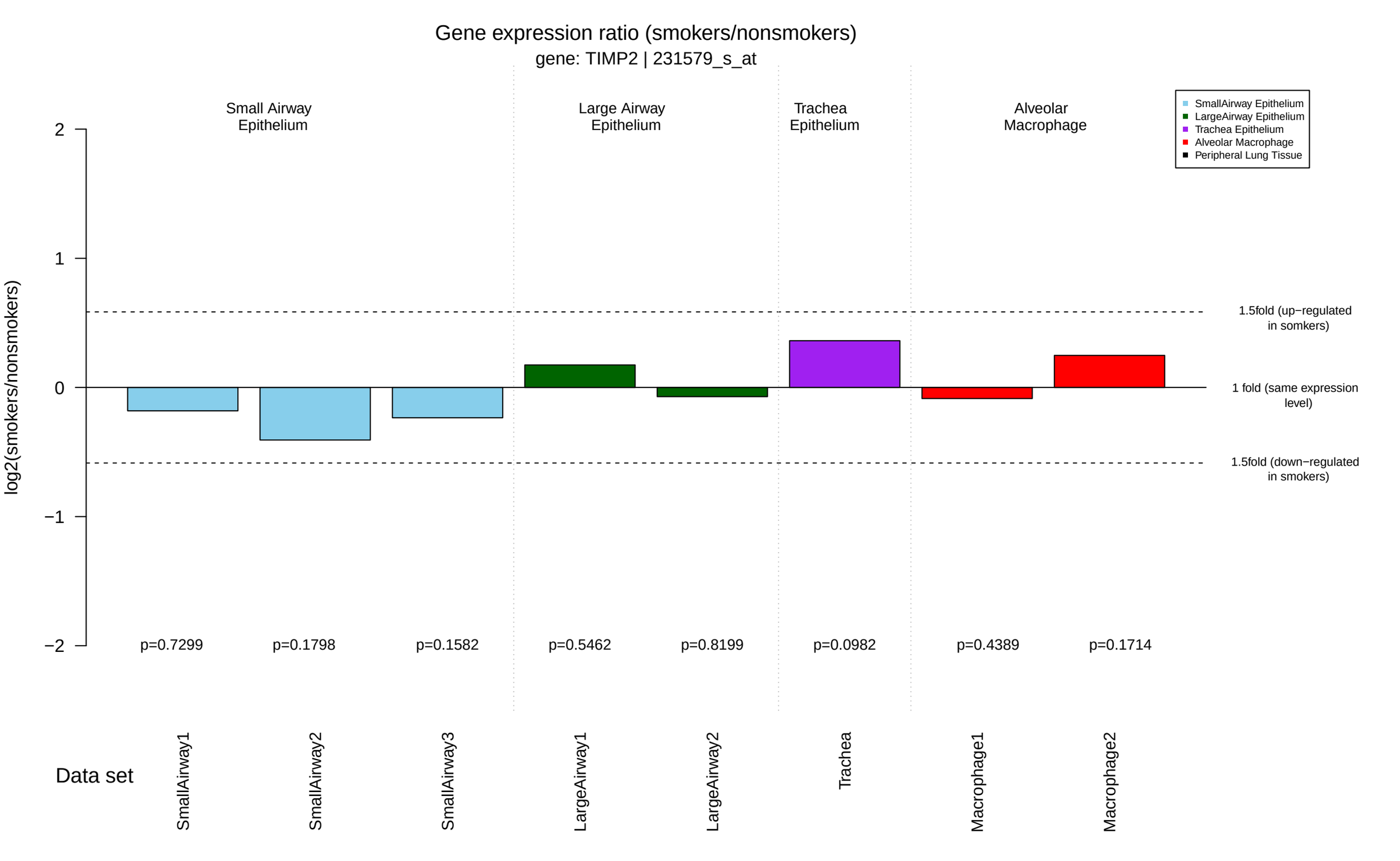
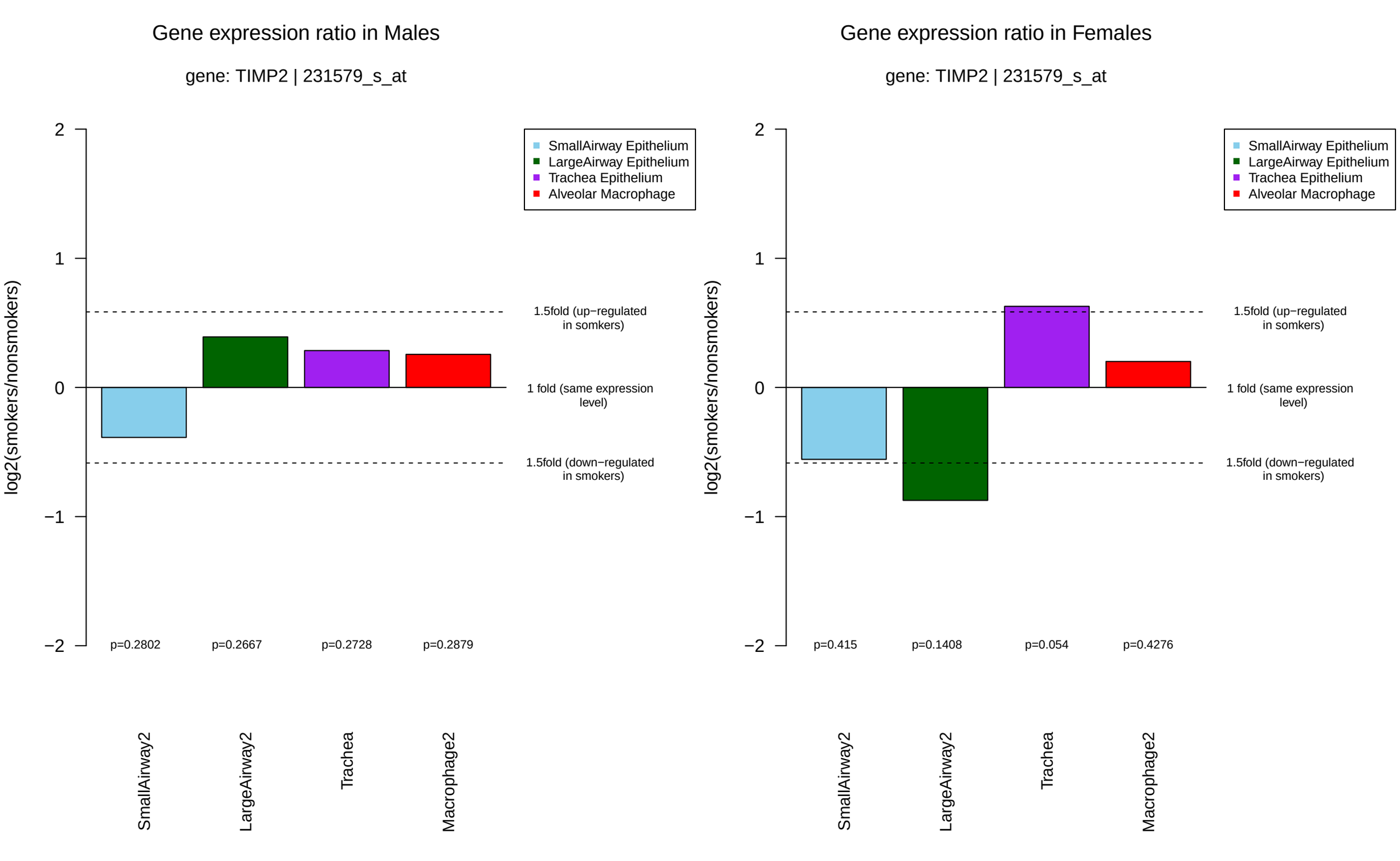
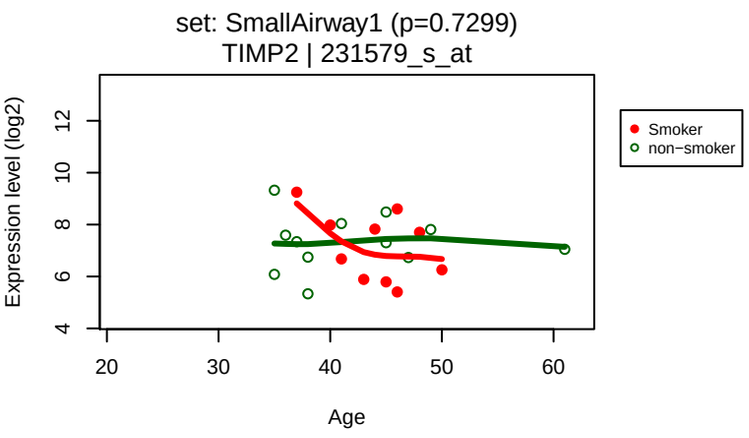
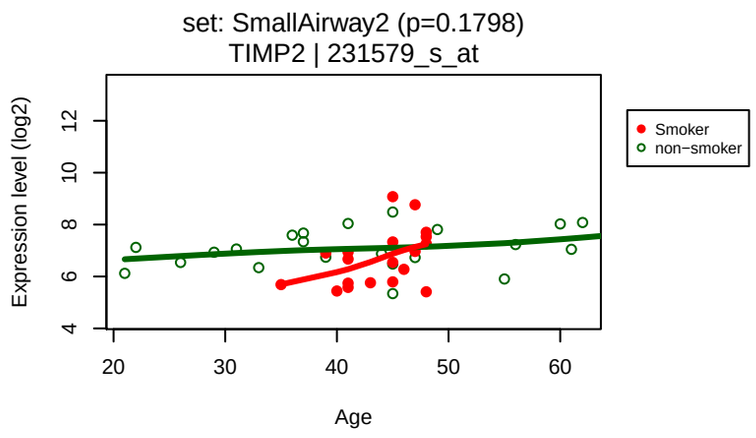
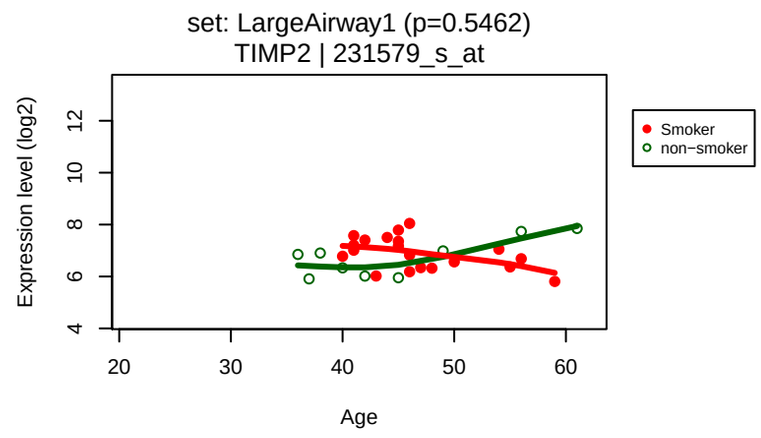

Gene expression of smokers and nonsmokers
gene: TIMP2 | 231579_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7299
p=0.1798
p=0.1582
p=0.5462
p=0.8199
p=0.0982
p=0.4389
p=0.1714
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

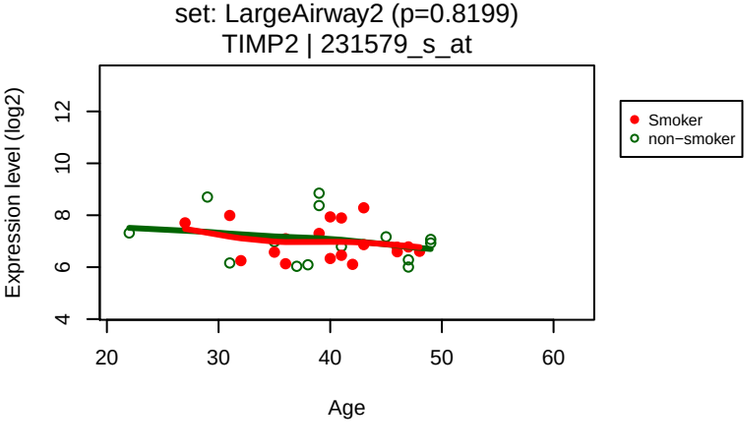

Gene expression ratio (smokers/nonsmokers)
gene: TIMP2 | 231579_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7299
p=0.1798
p=0.1582
p=0.5462
p=0.8199
p=0.0982
p=0.4389
p=0.1714
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

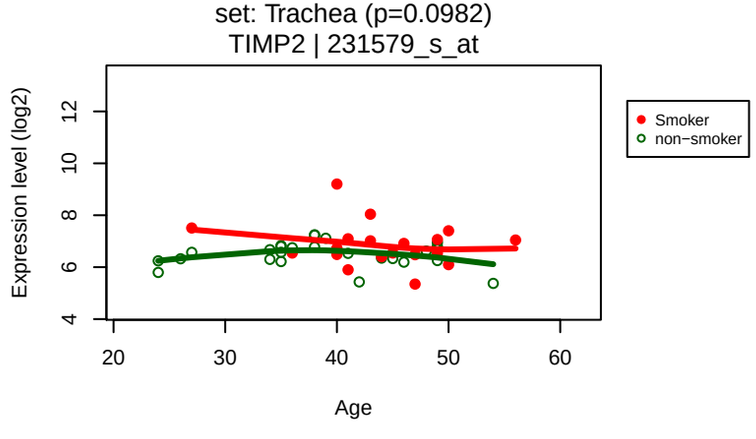

Gene expression ratio in Males
gene: TIMP2 | 231579_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2802
p=0.2667
p=0.2728
p=0.2879
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TIMP2 | 231579_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.415
p=0.1408
p=0.054
p=0.4276
SmallAirway2
LargeAirway2
Trachea
Macrophage2

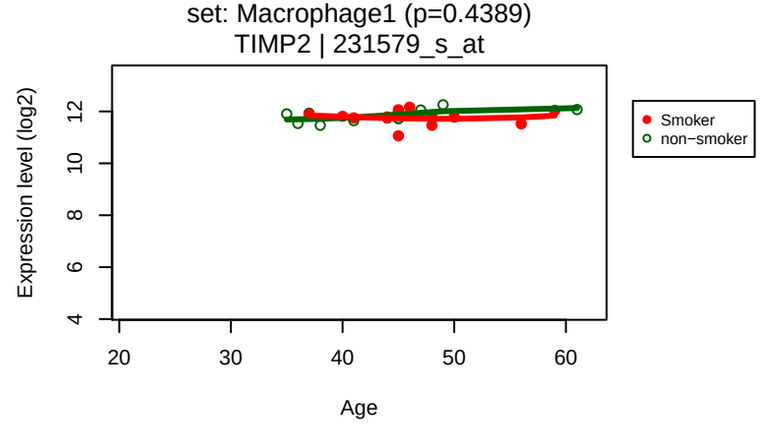

set: SmallAirway1 (p=0.7299)
TIMP2 | 231579_s_at
Expression level (log2)
Age
Smoker
non−smoker

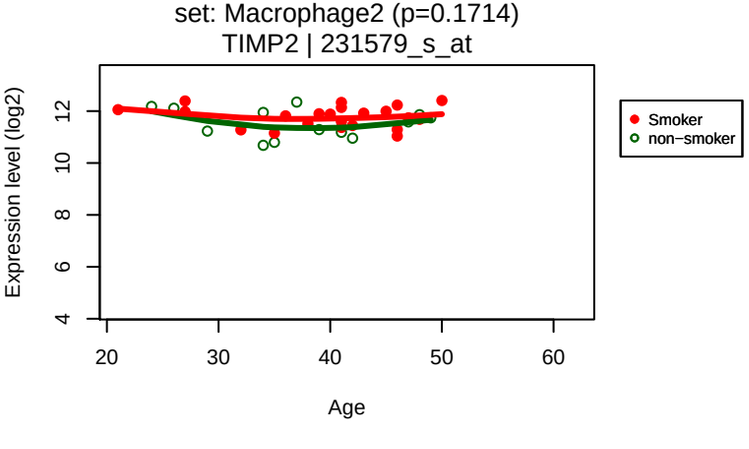

set: SmallAirway2 (p=0.1798)
TIMP2 | 231579_s_at
Expression level (log2)
Age
Smoker
non−smoker

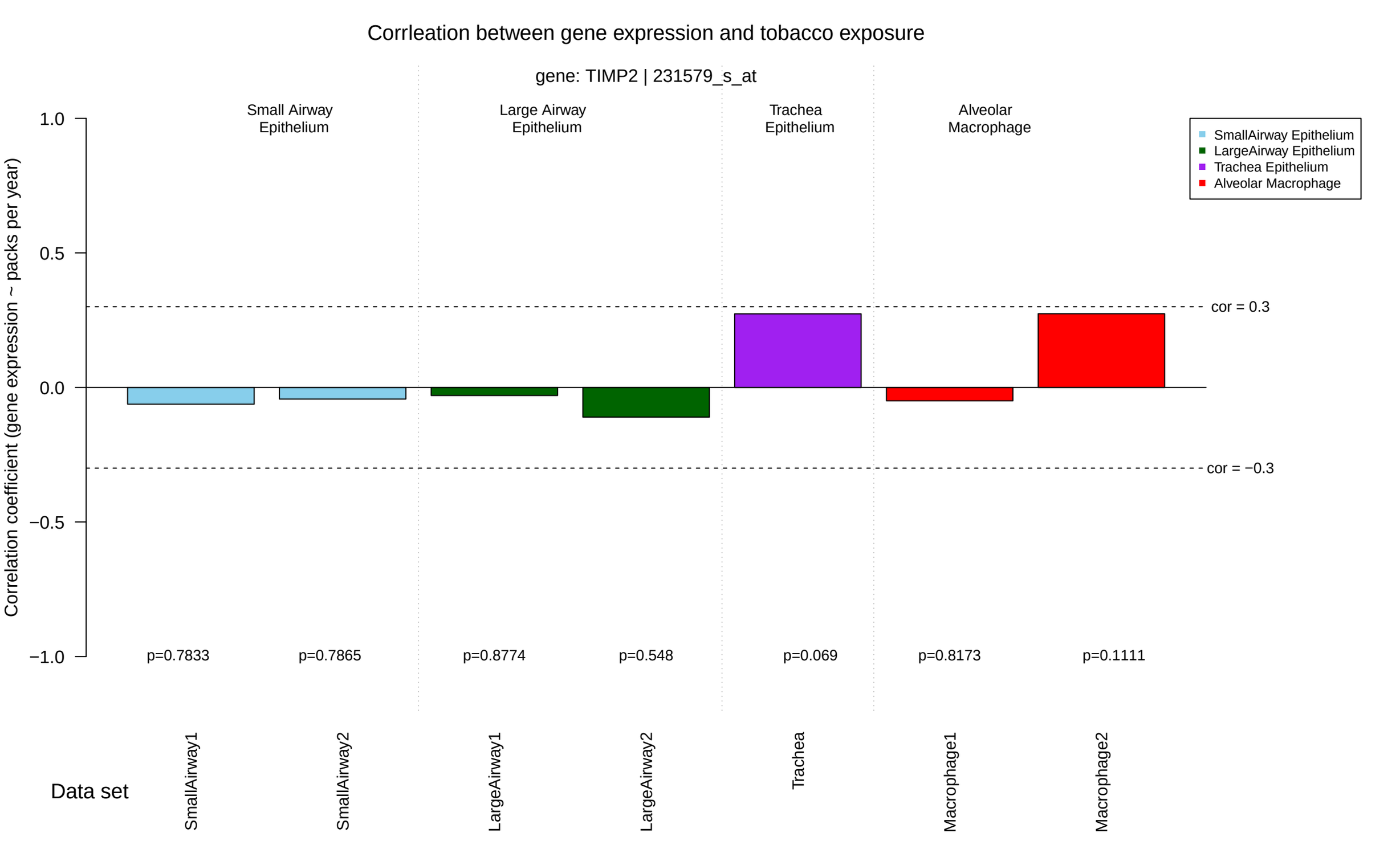

set: LargeAirway1 (p=0.5462)
TIMP2 | 231579_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8199)
TIMP2 | 231579_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0982)
TIMP2 | 231579_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4389)
TIMP2 | 231579_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1714)
TIMP2 | 231579_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TIMP2 | 231579_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7833
p=0.7865
p=0.8774
p=0.548
p=0.069
p=0.8173
p=0.1111
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2